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Loraine R. Tarou, Meredith J. Bashaw & Terry L. Maple

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Empty Nest: A Case Study of Maternal Response to Separation From a Juvenile Offspring in a Captive Sumatran Orangutan (*Pongo pygmaeus abelii*)

Loraine R. Tarou, Meredith J. Bashaw, and Terry L. Maple
TECHlab, Zoo Atlanta and School of Psychology
Georgia Institute of Technology

Improving the welfare of captive nonhuman primates requires evaluating the stressors created by the captive environment and reducing their negative effects. Social separation, although sometimes necessary for managing the genetic diversity of captive populations of animals, causes both psychological and physiological stress in human and primate monkey infants. Few studies have examined the maternal response of great ape mothers to separation from their offspring. This article describes the behavioral changes of a mother orangutan (*Pongo pygmaeus abelii)* after separation from her juvenile daughter. We collected data on measures of proximity and social behavior before the separation of the mother–infant dyad and of locomotion, arboreality, abnormal behavior, solitary behavior, and vocalization both before and after separation. We observed no behavioral indications of protest but observed some indications of despair after separation: decreased locomotion, increased inactivity, and increased self-directed behavior. In addition, we observed increases in arboreality and object-oriented behavior during morning sessions. These findings suggest that mother–juvenile separation in orangutans might be less stressful for mothers than might be expected. Such research has implications for the welfare and management of captive animals.

The social bond between mothers and their infants has been studied extensively in nonhuman primates, primarily by examining the behavioral and physiological responses of the infant to separation from the mother (Mineka & Suomi, 1978).

Requests for reprints should be sent to Loraine R. Tarou, TECHlab, Zoo Atlanta, 800 Cherokee Avenue Southeast, Atlanta, GA 30315. E-mail: ltarou@mindspring.com

Interest in this topic arose because both scientists were trying to establish a model for human depression and because the reaction of an animal to the removal of its social partners had implications for animal welfare (Mineka & Suomi, 1978). Bekoff, Gruen, Townsend, and Rollin (1992) emphasized the need for researchers to reevaluate continually their commitment to the ethical treatment of the animals they study. Maple (1995) wrote that zoos also must continue to improve the welfare of animals in their care. To improve the welfare of captive animals, we first must evaluate the effect of various stressors on the behavior of the animals and then attempt to minimize any negative effects of these stressors on welfare (Mason, 1991).

Social separation has been shown to influence welfare, affecting all four approaches to assessing well-being in nonhuman primates defined by Novak and Suomi (1988). These include immunosuppressive effects that compromise physical health (Laudenslager, 1988), corticosteroid increases that indicate a physiological stress response (Champoux & Suomi, 1994), and increased stereotypies and abnormal behaviors that might compromise both the behavioral repertoire and behavioral competence of the animal (Bard & Nadler, 1983). This article focuses on the behavioral response to separation.

Research with both humans (Bowlby, 1960) and other primates (Mineka & Suomi, 1978) has shown that infants exhibit a biphasic change in behavior after separation from their mothers. The first observed change is the protest phase, characterized by an increase in vocalization, locomotion, and stereotypic or abnormal behaviors after separation. A despair phase follows, characterized by a decrease below baseline levels of vocalization and locomotion and an increase in self-directed behavior. Behavioral responses similar to those observed in primate infants after maternal separation have been observed in nonrelated primate infants (Boccia, Reite, Kaemingk, Held, & Laudenslager, 1991), preadolescents and infants (Maple, Brandt, & Mitchell, 1975), juveniles (Bard & Nadler, 1983; Erwin, Mobaldi, & Mitchell, 1971; McKinney, Suomi, & Harlow, 1972), and even adults (Suomi, Eisele, Grady, & Harlow, 1975) after separation from their cagemates. Comparatively little information is available on the behavioral and physiological responses of the mother to separation from her infant.

In a review of behavioral pathology in primates, Capitanio (1986) discussed the lack of research on maternal response to infant separation and hypothesized that the maternal response would be similar to that observed in infants. However, available data suggest that the maternal response might be somewhat different from infant response. Studies of several species of monkey have reported behavioral indications of protest after separation from their infants but few or no signs of despair (Champoux & Suomi, 1994; Hinde & Davies, 1972; Jensen, 1968; Kaplan, 1970; Vogt & Levine, 1980). These behaviors often are accompanied by physiological stress, measured by increased levels of cortisol (Champoux & Suomi, 1994; Coe, Glass, Wiener, & Levine, 1983; Laudenslager, Held, Boccia, Reite, &

Cohen, 1990; Mendoza, Smotherman, Miner, Kaplan, & Levine, 1978; Vogt & Levine, 1980). The presence of both behavioral and physiological indications of stress suggests that separation might have implications for the welfare of captive nonhuman primates.

The great apes have been underrepresented in studies of mother–infant separation. The data available on the infant response to maternal separation indicate that great ape infants exhibit the characteristic biphasic change in behavior observed in other nonhuman primate species (Codner & Nadler, 1984; Hoff, Nadler, Hoff, & Maple, 1994). To our knowledge, there has been only one published study of the maternal response to infant separation in great apes. In a pilot study, Bloomsmith, Merhalski, and Gregor (1988) reported that mother chimpanzees (*Pan troglodytes*) exhibited very little evidence of protest or behavioral agitation and no signs of despair after the removal of their infants. In this study we report on the behavior of an orangutan mother after separation from her 6-year-old daughter.

METHOD

Subjects

The subject was a multiparous, wild-born, 42-year-old female orangutan. Before the separation, she and her 6-year-old daughter were housed at Zoo Atlanta in a 1515-m^2 outdoor exhibit from approximately 9:30 a.m. to 5:30 p.m. and in indoor holding facilities for the remainder of the time. For several weeks before the separation, the keeper staff trained the juvenile to leave her mother and enter the shipping crate for brief periods of time. Because of the age of the daughter and the crate training, neither animal required anesthesia to facilitate the separation. The orangutan Species Survival Plan requested the juvenile move to another zoo for breeding purposes. During the postseparation study period, the mother was housed alone in the same environment. The mother since has been introduced to, and paired with, another young female orangutan.

Data Collection

Behavioral data were collected for 24 days before and 16 days after the separation of the mother–infant dyad. We used an instantaneous scan sampling technique with 30-sec intersample intervals (Martin & Bateson, 1986) to record 21 behaviors representing six behavioral categories: locomotion, arboreality, abnormal behaviors, social behaviors, solitary behaviors, and vocalization (see Table 1). Proximity measures were used to determine the amount of time the mother and her daughter spent in contact with, in proximity to, and distant from each other before separation. All occurrence data were collected on vocalizations,

TABLE 1
Ethogram for the Study of Maternal Response to Separation in the Orangutan

Activity	
Locomote	Focal is moving from one place to another.
Stationary	Focal is sitting or lying.
Location in exhibit	
Arboreal	Focal has no limbs on the ground.
Terrestrial	Focal has one or more limbs on the ground.
Abnormal behaviors	
Pace	Focal is locomoting in a fixed pattern.
Regurge/reingest	Involves the regurgitation of food into mouth, hand or substrate, and subsequent reingestion of that food. Includes preregurgitation behaviors such as clearing a space for the regurgitation of food, shaking the head, twisting or grabbing the stomach, bending over, or rocking prior to regurgitation.
Feed regurge/reingest	Focal is eating the regurgitant of another animal.
Coprophagy	Focal is eating feces.
Feces manipulation	Focal is touching, smearing, or manipulating own or another's feces.
Urophagy	Focal is drinking urine.
Other	Focal is engaged in any abnormal behavior not included in the previous categories.
Social behaviors	
Social play	Focal is wrestling, play-biting, climbing, chasing or soliciting play with or from another individual. Often accompanied by play-face.
Other affiliative	Focal is engaged in grooming, social examination, or other social behavior that is not play.
Agonistic	Focal is engaged in either physical or nonphysical aggression toward another individual. Includes biting, hitting, displaying at, or supplanting another individual.
Solitary behaviors	
Feed	Focal is consuming primate chow, hay, straw, grass, browse, twigs, or leaves.
Drink	Focal is drinking water. Does not include drinking urine.
Object oriented	Focal is actively or passively touching, playing with, sniffing, or engaging with an object.
Self-directed	Includes scratching, picking at nose or skin, genital manipulation, or autogrooming.
Nest building	Focal is carrying or manipulating nesting material or building a nest.
Solitary play	Focal is engaged in play behavior that does not involve another animal.
Swing	Focal is swinging on arboreal substrate.
Other	Focal is engaged in any behavior not subsumed in the previous categories.
Vocalization	
Vocalization	Focal produces sound audible to observer.
Proximity	
Contact	Focal is physically touching another individual.
Proximate	Focal is within arm's length of another individual.
Distant	Focal is more than arm's length apart from another individual.

agonistic behaviors, and affiliative behaviors, which are often short in duration and might be underestimated by instantaneous sampling (Altmann, 1974). In addition to the behavioral and proximity measures, we also were interested in habitat use. The exhibit was divided into four areas. Areas 1–3 were visible to the public. Area 4 was not. During both the preseparation and postseparation phases of the study, data were collected in two consecutive 30-min sessions between 10:30 a.m. and noon and between 3:00 and 4:30 p.m. A total of 9.5 hr of baseline data was collected in the morning and 10 hr in the afternoon.

Data Analysis

Data were separated according to the phase of study (preseparation and postseparation) and time of day (morning and afternoon). Confidence intervals (CIs) were computed based on the mean and standard deviation for all behaviors in each behavioral category (Bard & Nadler, 1983). The percentage of scans in which each behavior was observed in the experimental phase was compared with the baseline data. Results from postseparation data that fell outside these CIs were considered to be statistically significant. For a more in-depth examination of postseparation behavior, the data from the first 4 days of the separation phase were separated from the last 12 days and compared with baseline. To compensate for computing multiple intervals, we used 99% CIs (Hays, 1994).

RESULTS

There were no behavioral indications of protest in either the morning or afternoon but some indications of despair in the afternoon. The mean percentage of scans the mother spent in locomotion did not change significantly in the postseparation phase in the morning. As depicted in Figures 1 and 2, arboreality increased significantly from 7.7% of scans in the preseparation phase to 33.2% of scans in the morning postseparation phase (CI: $0\% < X < 22\%$) and from 2.2 to 14.2% in the afternoon (CI: $0\% < X < 6\%$). Whereas arboreality increased in both the first 4 and last 12 days of the study in the afternoon, it increased only in the last 12 days of the study in the morning ($M = 43.1\%$; CI: $0\% < X < 22\%$).

There was no evidence of abnormal behavior in either the preseparation or postseparation phase. Of the solitary behaviors included in the ethogram, only feeding, object-oriented behavior, and self-directed behaviors could be analyzed, because the others did not occur with sufficient frequency. Feeding behavior did not change across the study. In the postseparation phase, object-oriented behavior increased dramatically from 3.9 to 30.5% of scans in the morning (CI: $0\% < X < 9\%$) and from 1.3 to 3.7% of scans in the afternoon (CI: $0\% < X < 3.3\%$, see Figure

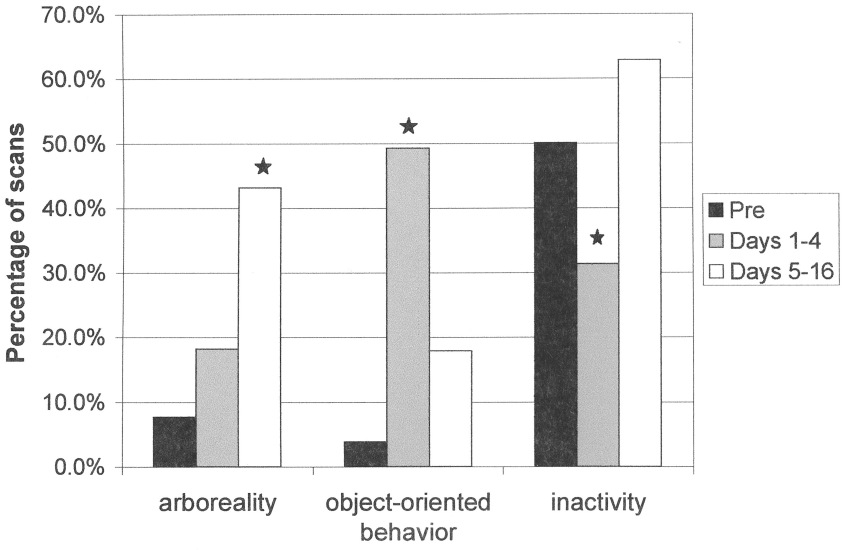


FIGURE 1 Significant changes in behavior in the morning as a function of separation phase. Stars indicate a value that is significantly different from baseline.

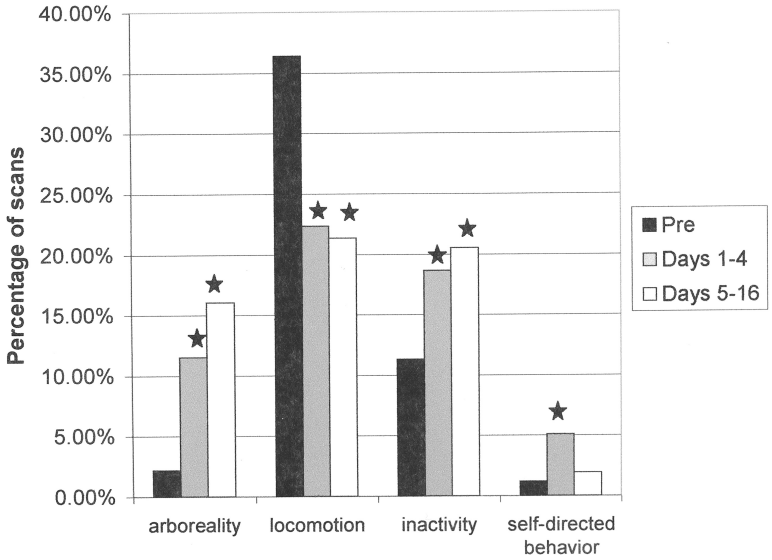


FIGURE 2 Significant changes in behavior in the afternoon as a function of separation phase. Stars indicate a value that is significantly different from baseline.

1). Further analysis showed that object-oriented behavior increased only in the first 4 days after separation ($M = 4.3\%$; CI: $0 < X < 3.3\%$). There was no change in self-directed behavior or locomotion in the morning, but there was a significant increase in self-directed behaviors in the afternoon ($M = 3.2\%$; CI: $0\% < X < 2.6\%$; see Figure 2). However, this afternoon change occurred only in the first 4 days after the separation ($M = 5.1\%$; CI: $0\% < X < 2.6\%$). Locomotion decreased in the afternoon ($M = 21.7\%$; CI: $27\% < X < 46\%$), and there was no significant difference between the first 4 days after separation and the last 12 days of the study. Finally, morning levels of inactivity decreased in the first 4 days after separation ($M = 31.4\%$; CI: $34\% < X < 66\%$). In contrast, inactivity increased in the postseparation phase in the afternoon ($M = 19.8\%$; CI: $3\% < X < 19.5\%$) but only during the last 12 days of the study ($M = 20.5\%$; see Figure 2).

Both behavior and habitat use differed between separation conditions. In the morning, the mean percentage of scans the subject spent in Area 3, which contained several climbing structures, increased significantly ($M = 62.7\%$; CI: $11\% < X < 62\%$). Further analysis showed that, as observed for arboreality, this increase occurred only during the last 12 days of the study ($M = 70.2\%$; CI: $11\% < X < 62\%$). In the afternoon, use of Area 2, which contained no climbing structures, decreased ($M = 10.5\%$; CI: $18\% < X < 35\%$), whereas use of Area 4, the area located closest to the indoor holding facilities, increased ($M = 38.7\%$; CI: $10\% < X < 30\%$). Further analysis indicated that there was a decrease in the use of Area 1 in the afternoon in the last 12 days of study ($M = 6.7\%$).

DISCUSSION

In this study, we observed significant changes in locomotion, arboreality, object-oriented behavior, self-directed behavior, inactivity, and habitat use in a mother orangutan after separation from her juvenile daughter. However, not all these changes indicated protest or despair. The protest phase generally is characterized by increases in vocalizations and locomotive behaviors, whereas despair is characterized by decreases in vocalization and activity and an increase in self-directed behaviors (Mineka & Suomi, 1978). We observed little evidence of protest but some evidence of behavioral despair in the orangutan mother after the separation of her juvenile daughter. The only behavioral change that could be considered indicative of protest was the decrease in inactivity seen in the morning sessions in the first 4 days after separation. This was not accompanied by increased vocalization, and locomotion actually decreased in the first 4 days after separation. Several changes were observed that traditionally have been associated with despair. In the postseparation period, locomotion decreased in the afternoon, self-directed behavior increased during the first 4 days after separation, and inactivity increased above baseline during the last 12 days of the study.

The few studies that have examined the maternal response to separation in monkeys found that mothers display signs of protest but relatively few or no signs of despair (Champoux & Suomi, 1994; Hinde & Davies, 1972; Jensen, 1968; Kaplan, 1970; Vogt & Levine, 1980). In the only published report of maternal response to separation in great apes, Bloomsmith et al. (1988) observed little evidence of protest and no sign of despair in chimpanzee mothers after the removal of their infants. The lack of observed signs of protest from the mothers after separation of their infant in this and the Bloomsmith et al. study might be due to the age of infants at time of separation. In most studies of the maternal response to infant separation in monkeys, the separated infant was younger than 8 months (Champoux & Suomi, 1994; Mendoza et al., 1978; Stanton & Levine, 1985; Vogt & Levine, 1980). In the Bloomsmith et al. study, the separated infants were between 21 and 23 months old, and in our study the separated offspring was 6 years old. In the wild, juvenile female orangutans disperse from their mothers at approximately 5–6 years of age (Tuttle, 1986). Possibly, the great ape mothers would have shown more signs of protest if their offspring had been separated at a younger age. Bloomsmith, Lambeth, Tarou, and Haberstroh (2000) tested this hypothesis with a sample of 15 chimpanzee mothers separated from their infants, including the mothers investigated in the previously mentioned pilot study. Again, the chimpanzee mothers showed little evidence of protest as indicated by a lack of change in locomotion and vocalizations and spent more time inactive after separation. There was no effect of age of infant, age of mother, presence of other offspring, or previous experience with mother–infant separation on the level of response.

The orangutan mother's despair response may have been more apparent in this study than in previous studies of maternal separation in great apes because she was not housed socially in the period immediately after separation. The mothers in the Bloomsmith et al. (2000) study remained in their social groups during the postseparation period, which might have attenuated their despair response. Although orangutans are predominantly solitary in the wild, in captivity they do well in social situations (Maple, 1980). For the orangutan mother, zookeepers might have become a substitute for the social opportunity no longer provided by the juvenile. Changes in behavior consistent with the operational definition of despair used here were observed only in the afternoon during the 2 hr before the mother was let inside for the evening. During these sessions, the mother spent more time close to the door of the indoor holding area (Area 4), either hanging on the mesh or sitting on the ground in front of the door to the indoor holding facilities. There also was a proportional decrease in time spent in the area farthest from the door (Area 2).

Whereas the majority of the behavioral changes traditionally associated with separation occurred during the afternoon, some of the more dramatic changes in other behaviors occurred during the morning observation sessions. For example, the time the mother orangutan spent engaged in object-oriented behavior in the morning dramatically increased after separation. The greatest increase occurred

during the first 4 days of the postseparation phase. Another interesting behavioral change was the time the mother spent arboreal. This behavior increased from 7.7% of the scans to 33.2% in the morning sessions and from 2.2% of the scans to 14.2% in the afternoon observation sessions. In the baseline period, the juvenile spent between 13 and 30% of her time engaged in these behaviors. Separation from the juvenile might have allowed more opportunity for the mother to engage in these behaviors, as the juvenile might have monopolized the objects and arboreal space provided for the pair before separation. In becoming more active and arboreal, the mother could have been responding in an adaptive manner by making herself more visible either to the "lost" juvenile or to an adult male with whom she could mate. Future studies should be designed to examine systematically these possible explanations for the mother's behavior.

In general, the removal of the juvenile from her mother affected the mother's behavior positively in some ways and negatively in others. Arboreality and object-oriented behavior increased after separation. These are species-typical behaviors usually considered to indicate an enriching environment and psychological well-being (Wilson, 1982). In the afternoon sessions, however, locomotion decreased and inactivity increased, which could be considered behavioral indications of compromised welfare. Because this is the first study to examine the maternal response to separation in the orangutan and because the study is limited to only one subject, it is unclear how representative the results are to the behavior of orangutan mothers in general. However, the results of this case study are similar to those of Bloomsmith et al. (1988)—who studied a larger sample of great ape mothers—suggesting that the maternal response to separation is different from the infant response to separation in great apes. This might have adaptive significance in that, from the infant's standpoint, remaining with the mother as long as possible is beneficial because she is a continuing source of food and information. However, from the mother's standpoint, a weaned infant directly competes for resources. There might be taxonomic differences in maternal response to separation, particularly between monkeys and great apes. In support of this argument, Maple and Hoff (1982) found that in the course of gorilla development, the infants, not the mothers, actively maintained proximity and contact between the pair. Future comparative studies using a greater number of subjects are needed to determine if there is a difference in maternal response to separation among monkeys, chimpanzees, and orangutans.

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REFERENCES

Altmann, J. (1974). Observational study of behavior: Sampling methods. *Behaviour, 48,* 227–265.

Bard, K. A., & Nadler, R. D. (1983). The effect of peer separation in young chimpanzees (*Pan troglodytes*). *American Journal of Primatology, 5,* 25–37.

Bekoff, M., Gruen, L., Townsend, S. E., & Rollin, B. E. (1992). Animals in science: Some areas revisited. *Animal Behavior, 44,* 473–484.

Bloomsmith, M. A., Merhalski, J. J., & Gregor, G. (1988). Chimpanzee (*Pan troglodytes*) mothers' response to separation from infants. *International Journal of Comparative Psychology, 2,* 57–61.

Bloomsmith, M. A., Lambeth, S. P., Tarou, L. R., & Haberstroh, M. D. (2000). *Maternal response to separation from offspring in chimpanzees (*Pan troglodytes*)*. Manuscript submitted for publication.

Boccia, M. L., Reite, M. L., Kaemingk, K., Held, P., & Laudenslager, M. L. (1991). Social context and reaction to separation in peer-reared pigtail macaques: Some preliminary observations. *Primates, 32,* 255–263.

Bowlby, J. (1960). Grief and mourning in infancy and early childhood. *Psychoanalytic Study of the Child, 15,* 9–52.

Capitanio, J. P. (1986). Behavioral pathology. In G. Mitchell (Ed.), *Comparative primate biology: Vol. 2. Behavior and ecology* (pp. 411–453). New York: Columbia University Press.

Champoux, M., & Suomi, S. J. (1994). Behavioral and adrenocortical responses of rhesus macaque mothers to infant separation in an unfamiliar environment. *Primates, 35,* 191–202.

Codner, M. A., & Nadler, R. D. (1984). Mother–infant separation and reunion in the great apes. *Primates, 25,* 204–217.

Coe, C. L., Glass, J. C., Wiener, S. G., & Levine, S. (1983). Behavioral, but not physiological, adaptation to repeated separation in mother and infant primates. *Psychoneuroendocrinology, 8,* 401–409.

Erwin, J., Mobaldi, J., & Mitchell, G. (1971). Separation of rhesus monkey juveniles of the same sex. *Journal of Abnormal Psychology, 78,* 134–139.

Hays, W. L. (1994). *Statistics.* Orlando, FL: Harcourt Brace College.

Hinde, R. A., & Davies, L. (1972). Removing infant rhesus from mother for 13 days compared with removing mother from infant. *Journal of Child Psychology and Psychiatry, 13,* 227–237.

Hoff, M. P., Nadler, R. D., Hoff, K., & Maple, T. L. (1994). Separation and depression in infant gorillas. *Developmental Psychobiology, 27,* 439–452.

Jensen, G. D. (1968). Reaction of monkey mothers to long-term separation from their infants. *Psychonomic Science, 11,* 171–172.

Kaplan, J. (1970). The effects of separation and reunion on the behavior of mother and infant squirrel monkeys. *Developmental Psychobiology, 3,* 43–52.

Laudenslager, M. L. (1988). The psychobiology of loss: Lessons from human and nonhuman primates. *Journal of Social Issues, 44,* 19–36.

Laudenslager, M. L., Held, P. E., Boccia, M. L., Reite, M. L., & Cohen, J. J. (1990). Behavioral and immunological consequences of brief mother–infant separation: A species comparison. *Developmental Psychobiology, 23,* 247–264.

Maple, T. L. (1980). *Orangutan behavior*. New York: Van Nostrand Reinhold.

Maple, T. L. (1995). Toward a responsible zoo agenda. In B. G. Norton, M. Hutchins, E. F. Stevens, & T. L. Maple (Eds.), *Ethics on the ark* (pp. 20–30). Washington, DC: Smithsonian Institute Press.

Maple, T. L., Brandt, E., & Mitchell, G. (1975). Separation of preadolescent from infant rhesus monkeys. *Primates, 16,* 99–101.

Maple, T. L., & Hoff, M. P. (1982). *Gorilla behavior*. New York: Van Nostrand Reinhold.

Martin, P., & Bateson, P. (1986). *Measuring behavior*. Cambridge, England: Cambridge University Press.

Mason, G. J. (1991). Stereotypies and suffering. *Behavioral Processes, 25,* 103–115.

McKinney, W. T., Suomi, S. J., & Harlow, H. F. (1972). Repetitive peer separations of juvenile-age rhesus monkeys. *Archives of General Psychiatry, 27,* 200–203.

Mendoza, S. P., Smotherman, W. P., Miner, M. T., Kaplan, J., & Levine, S. (1978). Pituitary–adrenal response to separation in mother and infant squirrel monkeys. *Developmental Psychobiology, 11,* 169–175.

Mineka, S., & Suomi, S. J. (1978). Social separation in monkeys. *Psychological Bulletin, 85,* 1376–1400.

Novak, M. A., & Suomi, S. J. (1988). Psychological well-being of primates in captivity. *American Psychologist, 43,* 765–773.

Stanton, M. E., & Levine, S. (1985). Brief separation elevates cortisol in mother and infant squirrel monkeys. *Physiology & Behavior, 34,* 1007–1008.

Suomi, S. J., Eisele, C. J., Grady, S. A., & Harlow, H. F. (1975). Depression in adult monkeys following separation from nuclear family environment. *Journal of Abnormal Psychology, 84,* 576–578.

Tuttle, R. H. (1986). *Apes of the world: Their social behavior, communication, mentality, and ecology*. Park Ridge, NJ: Noyes.

Vogt, J. L., & Levine, S. (1980). Response of mother and infant squirrel monkeys to separation and disturbance. *Physiology & Behavior, 24,* 829–832.

Wilson, S. F. (1982). Environmental influences on the activity of captive apes. *Zoo Biology, 1,* 201–209.